

Quantification of nitrate fate in a karst conduit using stable isotopes and numerical modeling

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ABSTRACT

Nitrate (NO_3^-) fate estimates in turbulent karst pathways are lacking due, in part, to the difficulty of accessing remote subsurface environments. To address this knowledge and methodological gap, we collected NO_3^- , $\delta^{15}\text{N}_{\text{NO}_3}$, and $\delta^{18}\text{O}_{\text{NO}_3}$ data for 65 consecutive days, during a low-flow period, from within a phreatic conduit and its terminal end-point, a spring used for drinking water. To simulate nitrogen (N) fate within the karst conduit, the authors developed a numerical model of NO_3^- isotope dynamics. During low-flow, data show an increase in NO_3^- (from 1.78 to 1.87 mg N L^{-1} ; $p < 10^{-4}$) coincident with a decrease in $\delta^{15}\text{N}_{\text{NO}_3}$ (from 7.7 to 6.8‰; $p < 10^{-3}$) as material flows from within the conduit to the spring. Modeling results indicate that the nitrification of isotopically-lighter ammonium ($\delta^{15}\text{N}_{\text{NH}_4}$) acts as a mechanism for an increase in NO_3^- that coincides with a decrease in $\delta^{15}\text{N}_{\text{NO}_3}$. Further, numerical modeling assists with quantifying isotopic overprinting of nitrification on denitrification (i.e., coincident NO_3^- production during removal) by constraining the rates of the two processes. Modeled denitrification fluxes within the karst conduit (67.0 ± 19.0 mg N m^{-2} d^{-1}) are an order-of-magnitude greater than laminar ground water pathways ($1\text{--}10$ mg N m^{-2} d^{-1}) and an order-of-magnitude less than surface water systems ($100\text{--}1000$ mg N m^{-2} d^{-1}). In this way, karst conduits are a unique interface of the processes and gradients that control both surface and ground water end-points. This study shows the efficacy of ambient N stable isotope data to reflect N transformations in subsurface karst and highlights the usefulness of stable isotopes to assist with water quality numerical modeling in karst. Lastly, we provide a rare, if not unique, estimate of N fate in subsurface conduits and provide a counterpoint to the paradigm that karst conduits are conservative source-to-sink conveyors.

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1. Introduction

Karst topography covers approximately 12% of the earth's surface and provides drinking water resources to nearly a quarter of the world's population (Ford and Williams, 2007). Karst landscapes are developed by the dissolution of chemically soluble bedrock leading to the formation of sinking streams, sinkholes, caves, and fracture networks (White, 2002). These dissolution features provide low-resistance pathways for water, sediment, and

contaminants, thus posing an environmental risk to downstream waterbodies. One such landscape, agricultural karst terrain, is particularly susceptible to contamination as recent anthropogenic activity has increased the availability of reactive nitrogen (N) and consequently the potential for eutrophication (Ford et al., 2019; Heffernan et al., 2012; Husic et al., 2019a, 2019b).

However, the fate of reactive N in karst sinking streams, caves, and conduits is lacking in the scientific literature. Previous studies neglect or marginalize N transformations in sinking streams and turbulent karst conduits (e.g., Mahler and Garner, 2009; McCormack et al., 2016; Perrin et al., 2007). These studies argue that because of high flow velocities karst conduits can be treated as conservative source-to-sink conveyors. Recent study has suggested

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that a reformulation of this conservative assumption is needed. First, epigenetic karst conduits often contain terrestrially derived sediment deposits and biofilms capable of mediating dissolved N transformations (Simon et al., 2007). Second, researchers have noted the presence of sediment-bound bacterial communities in karst caves and their potential to process terrestrially derived nitrate (NO_3^-), ammonium (NH_4^+), dissolved organic N (DON), sediment organic N (SON), and dinitrogen gas (N_2) (Heffernan et al., 2012; Henson et al., 2017). These bacterial communities can drive nitrification ($\text{NH}_4^+ \rightarrow \text{NO}_3^-$), mineralization ($\text{DON/SON} \rightarrow \text{NH}_4^+$), denitrification ($\text{NO}_3^- \rightarrow \text{N}_2$), and immobilization ($\text{NH}_4^+ \rightarrow \text{SON}$) reactions. The conservative transport assumption could lead to inaccurate estimates of the availability of reactive N, which may have worrisome consequences related to the occurrence of harmful algal blooms and the operation of municipal water systems.

Denitrification, as the permanent removal pathway of NO_3^- , affects ecosystem and biogeochemical cycles at local, regional, and global scales and is influenced by the spatiotemporal variability of ecological, hydrological, and geomorphological processes (Seitzinger et al., 2006). Primary drivers of denitrification include the availability of NO_3^- , organic carbon (C), and oxygen as well as particle size and hyporheic exchange (Boyer et al., 2006; Dähnke and Thamdrup, 2013). Given these drivers, rivers typically have the highest processing rates while groundwater pathways have the lowest (Seitzinger et al., 2006). In this way, karst conduits are a unique interface that combines processes and gradients that control surface stream and groundwater end-points. Despite these unique features, the net role of the conduit pathway towards removal is not well-known. The magnitude of NO_3^- removal in karst conduits relative to other pathways may serve as an indicator of the net role of karst topography in global N cycling and be of interest to water quality managers.

To quantify the aforementioned processes in turbulent karst conduits, collection of daily N samples from multiple points along the karst conduit seems a logical approach, and many previous studies in surface streams use this method (e.g., see review in Birgand et al., 2007); however, karst conduits, positioned many meters below the ground surface, can be difficult to locate, access, and instrument. For this reason, karst methods generally lag behind their surface counterparts (Hartmann et al., 2014). One advantage of the present study is that the primary conduit has already been located using electrical resistivity testing, at a location 5 km upstream of the spring, and was intersected by 3 groundwater wells, allowing for convenient sample collection (Zhu et al., 2011). The terminal resurgence of this conduit is at the largest spring in the Bluegrass Region of Kentucky where it serves as the primary municipal drinking water source for 13,000 people in Georgetown, KY. This study set-up provides an opportunity to collect unique data sets to assess the impact of in-conduit NO_3^- transformations on the drinking water source.

Ambient N stable isotopes have proven useful in surface and subsurface systems for separating sources of organic matter and nutrients (Husic et al., 2019a, 2017a), elucidating denitrification processes (Heffernan et al., 2012), and assisting with the reduction of uncertainty and equifinality in numerical models (Ford et al., 2017). Each N phase has a unique isotopic signature (δ), reported in units of per mil (‰) as the ratio of ^{15}N – ^{14}N in a sample versus a reference standard (e.g., $\delta^{15}\text{N} = [((^{15}\text{N}/^{14}\text{N})_{\text{sample}}/(^{15}\text{N}/^{14}\text{N})_{\text{air}}) - 1] \times 1000$) (Kendall et al., 2007). This isotopic signature is impacted by physical mixing and biogeochemical fate processes (Figure 1a). For example, as N cycles from one form to another, microbes prefer to utilize the isotopically lighter ^{14}N , which imparts a fractionation (ϵ) effect on the remaining substrate, leading to an enrichment in the substrate's $\delta^{15}\text{N}$ signature. Previous karst studies have used ambient N isotopes to infer the existence of denitrification

processes (e.g., Albertin et al., 2012; Heffernan et al., 2012); however, no studies, to our knowledge, use continuous daily sampling of $\delta^{15}\text{N}_{\text{NO}_3}$ and $\delta^{18}\text{O}_{\text{NO}_3}$ to estimate N transformations within a karst conduit.

The use of coupled elemental and isotopic data sets together with process-based numerical modeling shows promise for improving predictions of N fate and water quality in karst (Jensen et al., 2018). Numerical modeling can provide a continuous estimate of integrated processes within a study section provided inputs and outputs are measured and knowledge of hydrodynamic behavior in the testbed is known. Additionally, including stable isotope data streams adds an additional set of equations without additional unknowns and allows for the reduction of equifinality and improvement in model predictions (Ford et al., 2017). Further, numerical modeling of isotope dynamics allows for the estimation of the isotopic overprinting of nitrification on denitrification (i.e., coincident NO_3^- production during removal), which can better inform removal rates (Granger and Wankel, 2016). Literature exists for models of karst hydrology (Palanisamy and Workman, 2014), sediment transport (Husic et al., 2017b), and NO_3^- concentration (Ford et al., 2019). However, no studies to our knowledge have focused on numerical modeling of $\delta^{15}\text{N}_{\text{NO}_3}$ and $\delta^{18}\text{O}_{\text{NO}_3}$ fate in karst.

Overall, the objective of this study was to estimate the cycling and removal of N from within the sediment bed of a karst conduit. We collected 65 consecutive days of stable isotope ($\delta^{15}\text{N}_{\text{NO}_3}$ and $\delta^{18}\text{O}_{\text{NO}_3}$) data from within a subsurface conduit and at its terminal endpoint (a spring used for drinking water). Sample collection coincided with a two-month low-flow period during which internal cycling controlled N dynamics as surface-derived water (e.g. flow into sinkholes) was excluded from the study section. Thereafter, we constructed a numerical model coupling elemental and stable isotope mass balance equations and explored whether the integration of stable isotope data help to constrain model predictions and isotopic overprinting. We provide estimates of the rate of NO_3^- removal within karst conduits and provide a counterpoint to the paradigm that karst conduits are conservative source-to-sink conveyors.

2. Study site and materials

2.1. Study site

The Royal Spring groundwater basin (58 km²) drains part of the Cane Run watershed (96 km²) located in the Inner Bluegrass Region of Kentucky, USA (Figure 1b). The land surface is primarily agricultural in use (60%) with highly urbanized headwaters (40%) and a temperate climate (MAT: 13.0 ± 0.7 °C; MAP: 1170 $\pm$ 200 mm). The land surface is composed of moderately deep, well-drained soils underlain by phosphatic limestone of the Middle Ordovician period. The primary surface channel runs dry during 80% of the year due to flow pirating by swallets within Cane Run creek (Husic et al., 2017a). A turbulent phreatic cavern ($Re = 1 \times 10^5$), closely aligned with the main surface channel, supplies Royal Spring (243 m a.s.l.) with an average perennial discharge of $0.67 \text{ m}^3 \text{ s}^{-1}$. The Royal Spring aquifer serves as the raw municipal water source for the City of Georgetown, Kentucky. The urbanization of the uplands has resulted in bacteria and nutrient loadings that exceed standards set by the Clean Water Act and Kentucky Division of Water.

2.2. Materials

Two well-established sampling stations served to characterize water, sediment, and nutrient inputs (Phreatic Conduit) and outputs (Royal Spring) to the study section (Figure 1c). Due to the difficulty in locating subsurface conduits, the Phreatic Conduit is

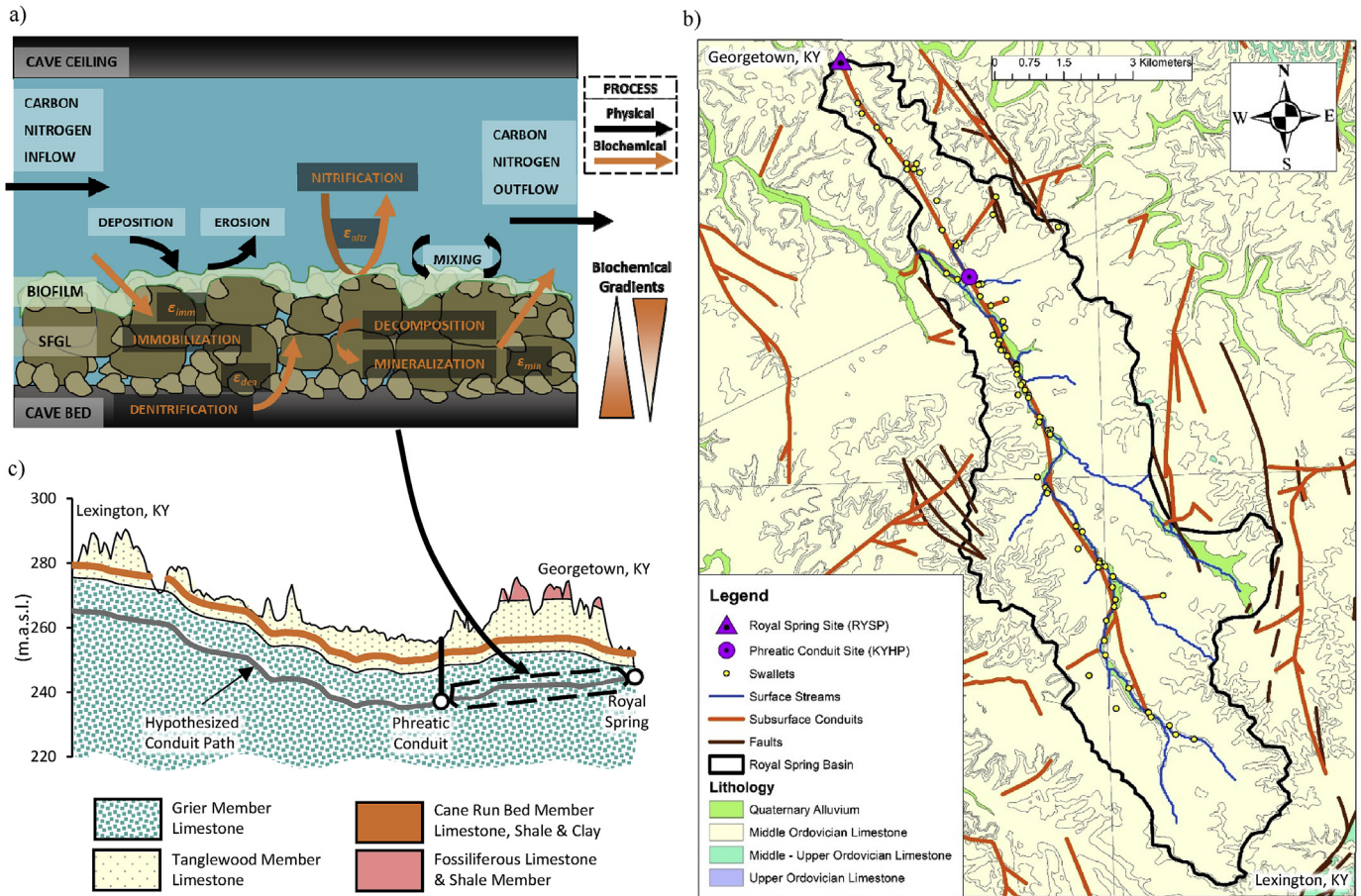


Fig. 1. (a) Conceptual model of physical and biogeochemical carbon and nitrogen processes in a subsurface conduit. Physical processes are shown in black arrows, and fractionation (ϵ) for each transformation is identified. (b) Cane Run watershed and Royal Spring basin indicating the location of the Phreatic Conduit and Royal Spring sites, surface stream, swallets, conduits, and faults. (c) Profile view of the hypothesized conduit path beneath the Cane Run watershed along with the superposition of geologic members. The dashed black box around a portion of the conduit path indicates the study section.

the only available location to serve as the upstream end of the study section. Based on dye tracing estimates, the percentage of the total Royal Spring basin drained at the Phreatic Conduit location is approximately 94% (Adams, 2019). The United States Geological Survey (USGS) operates a v-notch weir at Royal Spring (USGS 03288110). Previously published water and sediment data and modeling provide inputs to the numerical model developed in this study (Husic et al., 2019b, 2017a; 2017b). These inputs include the sediment exchange between the water column and karst bed, water and sediment fluxes into and out of the conduit, sediment particle size distributions entering and exiting the conduit, and the distribution of organic matter content and source material (i.e., soil, litter, and algae) entering the conduit. Sediment organic C elemental and isotopic mass balances were simulated in Husic et al. (2017b) and provide estimates of the longitudinal availability of organic matter to fuel N transformations. Sediment N elemental and isotopic mass balances were simulated in Husic (2018) and provide estimates of particulate N content and source material composition (i.e., soil, litter, and algae) as well as estimates of $\delta^{15}\text{N}$ associated with particulate matter in the watershed.

3. Methods

3.1. Collection and analysis of ambient N stable isotope data

In late summer 2017, over the course of 65 consecutive days, an

intensive field investigation measured physical, elemental, and isotopic data at two locations along the karst conduit. A well-level indicator (Slope 113583) recorded the depth to the water table at the Phreatic Conduit site. A multi-parameter probe (Horiba U-10) provided discrete temperature and specific conductivity measurements at both sites. At the Royal Spring site, we collected grab samples directly from the mouth of the spring into sterile 1 L bottles (I-Chem 312-0950BPC). At the Phreatic Conduit site, a deep well submersible pump (Hallmark Industries MA0414X-7) discharged water from the Phreatic Conduit to the surface for collection into 1 L bottles. To prevent biological activity from altering sample composition, we filtered 40 mL of sample through a 0.45 μm syringe filter (Whatman 6780–2504) and into a borosilicate vial with a permeable 1.5 mm septum (I-Chem TB36-0040). Samples remained refrigerated without the use of preservatives until elemental and isotopic analysis.

The Kentucky Geological Survey (KGS) laboratory analyzed NO_3^- samples following US EPA Method 300.0. Analysis featured a Dionex ICS-3000 Ion Chromatograph System with a carbonate-bicarbonate eluent generator and Dionex AS4A analytical column. The lab identified the NO_3^- concentration by comparing the retention time and peak area of the sample to a calibration curve generated from known standards. Lab ($n = 49$) and field ($n = 8$) duplicates had standard deviations of ± 0.02 and $\pm 0.07 \text{ mg N L}^{-1}$, respectively. No lab, field, or equipment blanks registered above the method detection limit. The University of Arkansas Stable Isotope

Laboratory (UASIL) produced isotopic data for NO_3^- with a Thermo Scientific GasBench II and Delta Plus IRMS using the bacterial denitrifier method (Casciotti et al., 2002). Reference standards used during N and O isotopic analysis include AIR and Vienna Standard Ocean Water (VSMOW), respectively. The isotopic reference materials for NO_3^- were USGS32 ($\delta^{15}\text{N}_{\text{NO}_3} = +180\text{‰}$, $\delta^{18}\text{O}_{\text{NO}_3} = -27.9\text{‰}$), USGS34 ($\delta^{15}\text{N}_{\text{NO}_3} = -1.8\text{‰}$), and USGS35 ($\delta^{18}\text{O}_{\text{NO}_3} = +57.5\text{‰}$). Duplicates of $\delta^{15}\text{N}_{\text{NO}_3}$ and $\delta^{18}\text{O}_{\text{NO}_3}$ ($n = 5$) had standard deviations of $\pm 0.28\text{‰}$ and $\pm 0.45\text{‰}$, respectively.

3.2. Isotope-aided numerical modeling of nitrate fate

3.2.1. Model conceptualization and background

We developed a numerical model to simulate fate and transport of dissolved N phases (i.e., NO_3^- , NH_4^+ , and DON) in a karst conduit (Fig. 2). This new model builds upon – and is coupled to – previously developed sediment C and N models for the study site (Husic, 2018; Husic et al., 2017a). The model runs at an hourly time-step at 1-km spatial increments, and was applied to the 5 km conduit reach between the Phreatic Conduit and Royal Spring sites. Given an average conduit flow velocity of $0.12 \pm 0.11 \text{ m s}^{-1}$ (Husic et al., 2017a), fluid starting at the Phreatic Conduit reaches Royal Spring within $10 \pm 9 \text{ h}$, providing ample time for down-gradient changes at the scale observed by our daily sampling routine (i.e. the model domain satisfies the CFL condition).

Daily field-collected data at the Phreatic Conduit, linearly interpolated from sample-to-sample at 1-h intervals, provided an upstream model boundary condition for NO_3^- , $\delta^{15}\text{N}_{\text{NO}_3}$, and $\delta^{18}\text{O}_{\text{NO}_3}$ inputs to the study section. We consider a 6% lateral inflow of water,

based on earlier dye tracing work (Adams, 2019), from the Phreatic Conduit to the Royal Spring. We parameterized the NO_3^- , $\delta^{15}\text{N}_{\text{NO}_3}$, and $\delta^{18}\text{O}_{\text{NO}_3}$ of this inflow as the mean ± 2 standard deviations of all low-flow data collected during this study as the land use distribution (i.e. horse farms and hay pasture) in the lower 6% of the spring basin is the same as the distribution in the upper 94%. Both a water budget ($Q_{\text{RS}} = 0.99Q_{\text{PC}}$; $R^2 = 0.77$; Husic, 2018) and stable isotope comparison ($\delta^2\text{H}_{\text{H}_2\text{O}}$ and $\delta^{18}\text{O}_{\text{H}_2\text{O}}$; Husic et al., 2019a) indicate a relatively small amount of lateral inflow during high flows and we assume this relationship also holds for low flows. Lastly, we evaluate model performance by comparing measured NO_3^- and $\delta^{15}\text{N}_{\text{NO}_3}$ values at Royal Spring with model-simulated results.

3.2.2. Model Equations and formulation

Many physical and biogeochemical processes affect NH_4^+ , DON, NO_3^- , $\delta^{15}\text{N}_{\text{NO}_3}$, and $\delta^{18}\text{O}_{\text{NO}_3}$ mass balances (Fig. 2). The mass balance of dissolved N species (kg N) within a cell is a function of upstream inflow, downstream outflow, lateral inflow, and biogeochemical processing of N facilitated by bed sediment, and was modeled as

$$N_{(i)}^{(j)(n)} = N_{(i-1)}^{(j)(n)} + \left[Q_{(i-1)}^{(j-1)(n)} C_{(i-1)}^{(j-1)(n)} - Q_{(i)}^{(j)(n)} C_{(i)}^{(j)(n)} \right] \Delta t + \left[Q_{\text{lat}(i)}^{(j)(n)} C_{\text{lat}(i)}^{(j)(n)} \right] \Delta t \pm R_{(i)}^{(j)(n)} \quad (1)$$

where i is the model time step; j is the model spatial step; n is the index for dissolved N-species (i.e. NO_3^- , NH_4^+ , DON); $N_{(i-1)}^{(j)(n)}$ is the mass of N from the previous time step (kg N); $Q_{(i-1)}^{(j-1)(n)}$ and $Q_{(i)}^{(j)(n)}$ are the flow rate into and out of the cell ($\text{m}^3 \text{ s}^{-1}$), respectively;

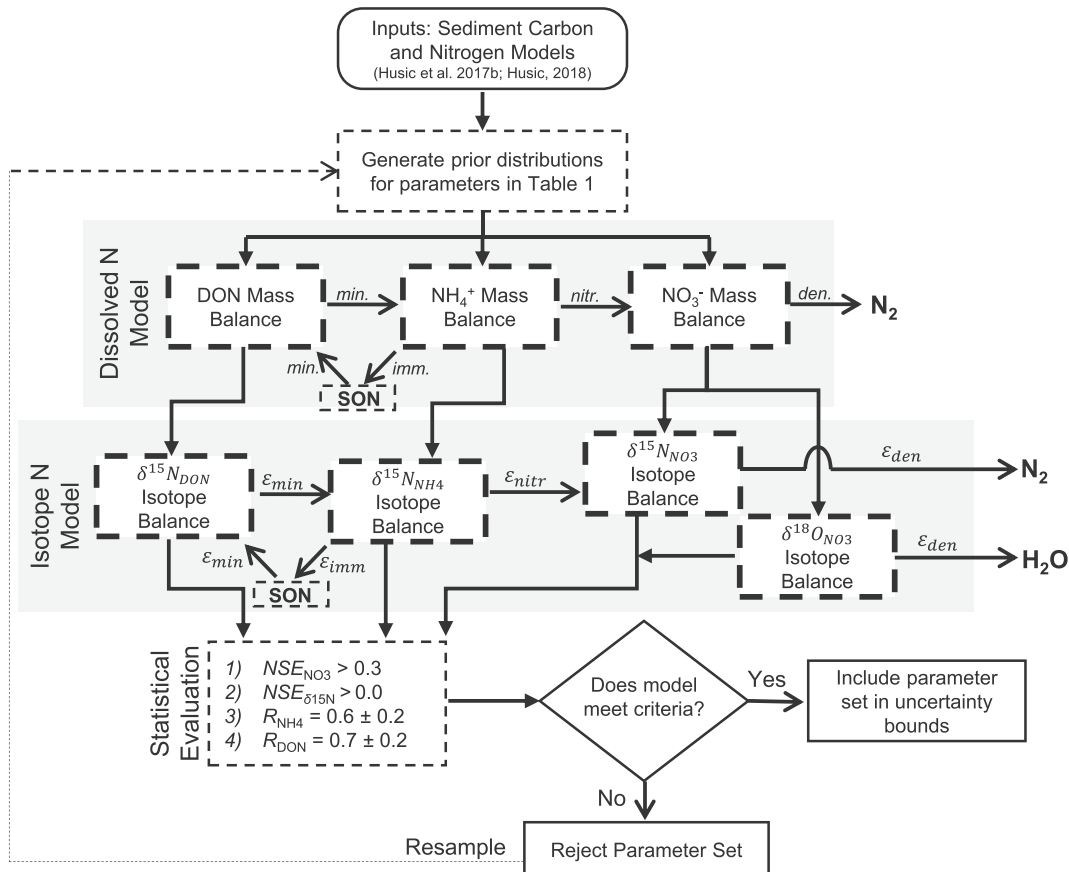


Fig. 2. Model calibration and uncertainty framework for evaluating NO_3^- , $\delta^{15}\text{N}_{\text{NO}_3}$, NH_4^+ , and DON model results. NSE_{NO_3} , $NSE_{\delta^{15}\text{N}}$, R_{NH_4} and R_{DON} are defined in Section 3.2. For a definition of other terms see Table 1.

$C_{(i-1)}^{(j)(n)}$ and $C_{(i)}^{(j)(n)}$ are the concentration of an N species flowing into and out of the cell (mg N L^{-1}), respectively; $Q_{\text{lat}(i)}^{(j)(n)}$ is the lateral inflow rate into the cell from the surrounding bedrock ($\text{m}^3 \text{s}^{-1}$), $C_{\text{lat}(i)}^{(j)(n)}$ is the concentration of an N species flowing laterally into the cell (mg N L^{-1}), Δt is the model time step (i.e. 1 h) (s); and $R_{(i)}^{(j)(n)}$ is the biogeochemical processing of N by bed sediment that can contribute to or remove from an N phase (kg N) and is defined in detail in Table S1.

In cave systems, 99% of microorganisms reside within fine sediment (Lehman et al., 2001), thus we assume that transformations of the dissolved N pool are primarily facilitated by bacteria inhabiting the karst conduit bed (see Fig. S2 for downhole imagery of a thin sediment mat blanketing the conduit bed). The mass balance of sediment organic N (SON) (kg N) is composed of three constituent pools (in order of lability: algae, litter, and soil; Husic et al., 2017a) and was modeled as

$$S_{(i)}^{(j)(k)} = S_{(i-1)}^{(j)(k)} - E_{(i)}^{(j)(k)} + D_{(i)}^{(j)(k)} \pm X_{(i)}^{(j)(k)} \pm R_{(i)}^{(j)(n)} \quad (2)$$

where k is an index for N-source (i.e. algae, litter, soil), $S_{(i-1)}^{(j)(k)}$ is the supply of sediment N from the previous time step (kg N), $E_{(i)}^{(j)(k)}$ is the amount of sediment N eroded (kg N), $D_{(i)}^{(j)(k)}$ is the amount of sediment N deposited (kg N), and $X_{(i)}^{(j)(k)}$ is the equilibrium exchange of N between suspended and bed sediment (kg N). A thorough description of sediment, particulate C, and particulate N transport modeling and equation coupling is detailed in Husic (2018).

Coupling of the dissolved and particulate pools of N is facilitated by transformations in the sediment bed. The mass of N exchanged between all pools (kg N) was modeled as

$$R_{(i)}^{(j)(n)} = \pm N_{\text{min}(i)}^{(j)(n)(d)} \pm N_{\text{nitr}(i)}^{(j)(n)} \pm N_{\text{imm}(i)}^{(j)(n)} \pm N_{\text{den}(i)}^{(j)(n)} \quad (3)$$

where d is an index for organic N phase (i.e. DON or SON), $N_{\text{min}(i)}^{(j)(n)(d)}$ is the amount of organic N mineralized to NH_4^+ (kg N), $N_{\text{nitr}(i)}^{(j)(n)}$ is the mass of NH_4^+ oxidized to NO_3^- in the water column (kg N), $N_{\text{imm}(i)}^{(j)(n)}$ is the mass of N immobilized into SON by biota (kg N), and $N_{\text{den}(i)}^{(j)(n)}$ is the mass of NO_3^- denitrified to N_2 (kg N). Sorption is not considered as low flows are unlikely to agitate the conduit bed and release stored NH_4^+ or NO_3^- . Equations for each of these reactions are shown in the Supplemental Information (Table S1) and their respective terms are defined therein. Finally, only the relevant transformations, enumerated in Equation (3), are applied to each N pool. For example, if considering mass balance changes to the DON pool, the denitrification and nitrification terms would effectively equal "0" as they do not directly impact the mass of DON.

The cycling of N between oxidation states is recognized to discriminate in favor of lighter isotopes in a process termed fractionation (Kendall et al., 2007). The isotopic mass balance (‰) accounting for mixing of sources and biogeochemical transformations through space and time was modeled as

$$\delta_{(i)}^{(j)(n)} = \delta_{(i-1)}^{(j)(n)} X_{(i-1)}^{(j)(n)} + \sum \delta_{\text{inputs}(i)}^{(j)(n)} X_{\text{inputs}(i)}^{(j)(n)} - \sum \delta_{\text{outputs}(i)}^{(j)(n)} X_{\text{outputs}(i)}^{(j)(n)} - \sum \epsilon \ln(f_{(i)}^{(j)(n)}) \quad (4)$$

where $\delta_{(i)}^{(j)(n)}$ is the isotopic signature of a given pool (‰); $X_{(i)}^{(j)(n)}$ is the fraction of an element in a given pool; $\delta_{\text{inputs}(i)}^{(j)(n)}$ is the isotopic signature of an input (‰) and $X_{\text{inputs}(i)}^{(j)(n)}$ is the fraction of an element in the input (e.g., upstream and lateral inflows); $\delta_{\text{outputs}(i)}^{(j)(n)}$ is the isotopic signature of an output (‰) and $X_{\text{outputs}(i)}^{(j)(n)}$ is the fraction of an element in the outflowing material; ϵ is the enrichment factor for a process (and is simulated using a Rayleigh-type model; Kendall et al., 2007) (‰); and $f_{(i)}^{(j)(n)}$ is the fraction of remaining substrate.

3.2.3. Model parameterization, evaluation, and uncertainty

Model calibration parameters were bound by field-collected data and literature-derived values (Table 1). Uncertain inputs were estimated using Monte Carlo sampling from uniform distributions (Fig. 2). Stable isotope enrichment factors (ϵ) and coefficients for biogeochemical reactions (β and k) were derived from published work on C and N cycling in human-impacted streams (Ford et al., 2017; Kendall et al., 2007; Ryzhakov et al., 2010). Based on historic data collected in the Cane Run watershed, we parameterized ranges for NH_4^+ concentration (C_{NH_4}), DON concentration (C_{DON}), and DON–N stable isotope composition ($\delta^{15}\text{N}_{\text{DON}}$) for water recharging the conduit (Husic, 2018; Husic et al., 2019b). Using an aggregated range reported by Kendall et al. (2007), we placed bounds on the possible values of the NH_4^+ –N stable isotope composition ($\delta^{15}\text{N}_{\text{NH}_4}$) of conduit recharge.

Forty-four of the 65 daily samples occurred during low-flow periods, defined as 10 or more consecutive days of baseflow (i.e., $<0.3 \text{ m}^3 \text{s}^{-1}$). These 44 samples provided the data for statistical model evaluation. For normally distributed data, a two-tailed t -test ($\alpha = 0.05$) was used to assess differences between upstream and downstream constituent composition. For non-normal data, the non-parametric Mann-Whitney Rank Sum test ($\alpha = 0.05$) was used instead. We evaluated normality using the Kolmogorov-Smirnov test. The NO_3^- and $\delta^{15}\text{N}_{\text{NO}_3}$ data sets were randomly divided into calibration and validation sets of equal cardinality. We evaluated NO_3^- and $\delta^{15}\text{N}_{\text{NO}_3}$ model performance using the Nash-Sutcliffe Efficiency (NSE):

$$\text{NSE} = 1 - \frac{\sum_{t=1}^T (Q_m^t - Q_o^t)^2}{\sum_{t=1}^T (Q_m^t - \bar{Q}_o)^2} \quad (5)$$

where T is the total number of observations, Q_o^t is the observed value at time t , Q_m^t is the modeled value at time t , and $\bar{Q}_o$ is the mean of all observed values. The NSE ranges from $-\infty$ to 1, with 1 indicating a perfect match of modeled results to data and 0 indicating that the model provides no more predictive capability than the mean of observed data (Moriassi et al., 2007). The minimum objective criteria for modeled NO_3^- (NSE $_{\text{NO}_3}$) and $\delta^{15}\text{N}_{\text{NO}_3}$ (NSE $_{\delta^{15}\text{N}}$) were set at 0.3 and 0.0, respectively. NSE $_{\delta^{18}\text{O}}$ was not considered as the data do not show a discernible trend. While minimum objective criteria for water and sediment are commonly reported (e.g. Moriassi et al., 2007), criteria for NO_3^- and $\delta^{15}\text{N}_{\text{NO}_3}$ are not well-established, therefore the authors impose that the NO_3^- and $\delta^{15}\text{N}_{\text{NO}_3}$ models should be better than (>0.3) and at least as good (>0.0) as the mean of the data, respectively. We further constrained model results such that the ratio (R) of modeled DON and NH_4^+ concentrations exiting-vs-entering the conduit, R_{DON} and R_{NH_4} , respectively, fall within the bounds of observed data. From a historical dataset, the means of these ratios are 0.7 and 0.6 (see Fig. S1 for data), respectively, and, due to data-scarcity, model simulations

Table 1
Calibration parameters for the elemental and isotopic NO_3^- fate and transport model, including parameter description, simulated and calibrated ranges, units, and reference(s).

Parameter	Description	Range Simulated	Range Calibrated ^a (Pre-isotope)	Range Calibrated ^a (Post-isotope)	Units	Reference(s)
β_{den}	Coefficient for denitrification reaction	$5 \times 10^{-10} - 5 \times 10^{-7}$	$9 \times 10^{-10} - 6 \times 10^{-8}$	$4 \times 10^{-9} - 7 \times 10^{-8}$	$[\text{kg N m}^{-2} \text{ s}^{-1}]$	Mulholland et al., 2008, Ford et al., 2017
k_{min}	First-order DON mineralization constant	0–0.20	0.06–0.19	0.04–0.19	$[\text{d}^{-1}]$	Zhang et al., 2017, Ryzhakov et al. 2010
k_{nitr}	First-order nitrification constant	0–0.68	0.14–0.65	0.12–0.50	$[\text{d}^{-1}]$	
$C_{\text{DON-rec}}$	DON concentration of recharge	0.20–0.50	0.21–0.49	0.20–0.48	$[\text{mg N L}^{-1}]$	Measured at site; Husic, 2018; Husic et al., 2019b
$C_{\text{NH}_4\text{-rec}}$	NH_4^+ concentration of recharge	0–0.60	0.07–0.55	0.32–0.59	$[\text{mg N L}^{-1}]$	
ϵ_{nitr}	Nitrification isotope enrichment factor	1–26	N/A	1.12–10.0	[–]	Kendall et al., 2007, Ford et al., 2017
ϵ_{imm}	Immobilization isotope enrichment factor	1–13	N/A	1.19–12.7	[–]	
ϵ_{den}	Denitrification isotope enrichment factor	1–18	N/A	1.05–7.69	[–]	
$\delta^{15}\text{N}_{\text{DON-rec}}$	$\delta^{15}\text{N}_{\text{DON}}$ composition of conduit recharge	2–9	N/A	2.10–8.79	[‰]	Measured at site; Husic, 2018
$\delta^{15}\text{N}_{\text{NH}_4\text{-rec}}$	$\delta^{15}\text{N}_{\text{NH}_4}$ composition of conduit recharge	–10 – +5	N/A	–9.97––4.79	[‰]	Kendall et al., 2007

^a Range calibrated includes 95% of the modeled prediction interval.

falling within ± 0.2 of these means were accepted during model evaluation (Husic et al., 2019b).

Further, to assess the presence and impact of isotopic overprinting of nitrification on denitrification, we ran several scenarios through the calibrated numerical model. We tested assumptions regarding nitrification, denitrification, and conservative transport and assessed the impact on model statistics when including biogeochemical reactions vs excluding them. The three scenarios were: 1) no denitrification, 2) no nitrification, and 3) conservative transport (i.e. all reaction rates set to “0”).

Lastly, we assessed model uncertainty and parameter sensitivity before and after integrating stable isotope data using the Generalized Likelihood Uncertainty Estimation (GLUE) methodology (Beven, 2006). GLUE is initiated by assuming a prior distribution for model parameters, simultaneously sampling all parameters, and then retaining only parameter sets that satisfy the minimum objective criteria (Husic et al., 2019b). A posterior distribution can then be constructed from the set of acceptable evaluations, and uncertainty bounds can be generated for NO_3^- and $\delta^{15}\text{N}_{\text{NO}_3}$ predictions. A total of 5.5×10^6 simulations were performed. Uncertainty analysis was performed on an institutionally shared high-performance computing cluster with 9240 processor cores and 57 TB of total RAM.

3.3. Nitrate removal in sinking streams

To provide a comparison of NO_3^- removal estimates in karst sediment bed to other freshwater systems, we compiled published denitrification and nitrification rates. Studies include other karst aquifers (Heffernan et al., 2012), karst lakes (McCormack et al., 2016), non-karst lakes (Piña-Ochoa and Álvarez-Cobelas, 2006), streams from diverse biomes (Arango et al., 2007), agricultural soil and groundwater (Anderson et al., 2014; Jahangir et al., 2012), and global estimates for groundwater and river sediment (Seitzinger et al., 2006). The Heffernan et al. (2012) study investigated 61 springs in the Upper Floridian Aquifer (USA) and estimated that, despite relatively low rates, denitrification accounted for the removal of a considerable fraction of aquifer N inputs. McCormack et al. (2016) calculated denitrification in karst turloughs (disappearing lakes) by a mass-balance calculation. The study by Arango et al. (2007) covers numerous streams, including those impacted by urban, agricultural, and forested land uses. The studies by Piña-Ochoa and Álvarez-Cobelas (2006) and Seitzinger et al.

(2006) aggregate many different denitrification estimates from all over the world. Only estimates made from within the sediment (i.e. not the water column) were used in calculations for this study. Lastly, denitrification in riparian soils and deep groundwater are also estimated from numerous sites (Anderson et al., 2014; Jahangir et al., 2012). We also aggregate studies of nitrification, including prairie and agriculture-impacted streams (Kemp and Dodds, 2002) and open-channel karst caverns (Simon and Benfield, 2002).

4. Results and discussion

4.1. Collection of ambient N stable isotope data

During the 65 days of field sampling, three low-flow periods, defined as 10 or more consecutive days of baseflow, were identified (Figure 3a). The elevation of water in the Phreatic Conduit observation well was only slightly (< 0.5 m) above the springhead elevation, providing energy to drive flow downstream (Figure 3b). Temperature in the conduit decreased down-gradient due to the relatively low background temperature of the aquifer ($T_{\text{ROCK}} \sim 13$ °C). The differences in temperature were statistically significant ($p < 10^{-3}$) and can be accounted for by thermal convection at the conduit wall ($T_{\text{ROCK}} - T_{\text{H}_2\text{O}} \approx -4$ °C). In terms of specific conductance, potential ion exchange of water with the limestone bedrock caused longitudinal increase in conductivity (from 601 $\mu\text{S}/\text{cm}$ at Phreatic Conduit to 608 $\mu\text{S}/\text{cm}$ at Royal Spring), but the increase was not statistically significant ($p = 0.83$). This result was not surprising as other modeling studies in karst indicate that the conductivity response during transport changes half as slowly as temperature (Winston and Criss, 2004). Data trends in water, temperature, and conductivity provide circumstantial evidence that the study section of the phreatic conduit represents a relatively closed system during low-flow.

Nitrate and ambient N stable isotope data indicate that the karst sediment bed acts as a net source of NO_3^- during transport (Fig. 4). While upstream inputs of NO_3^- decreased by an average of 1–5% per day, the mean concentration of NO_3^- transported between the Phreatic Conduit and Royal Spring increased by an average of 5%, from 1.78 to 1.87 mg N L^{-1} ($p < 10^{-4}$) (Figure 4a, b). The biogeochemical fate of NO_3^- in the conduit during the study period appears to be dominated by production processes rather than removal. Potential reasons for an increase in NO_3^- concentration could be from the nitrification of NH_4^+ , the mineralization of organic

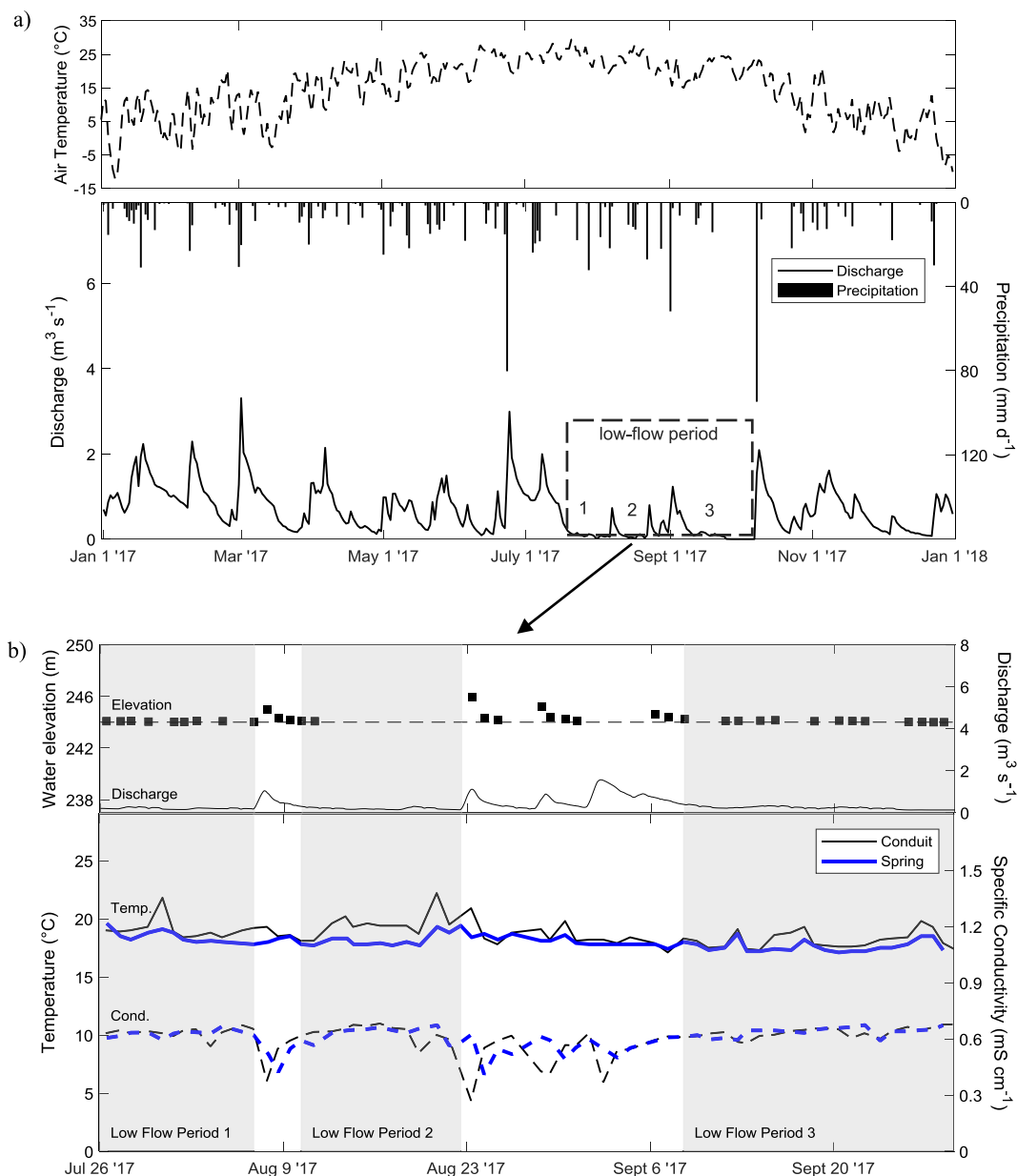


Fig. 3. (a) Daily average air temperature, spring discharge, and precipitation intensity for the 2017 calendar year. The numbers inset in the dashed box enumerate the low-flow events contained within a broader 2-month dry period. (b) The top sub-plot shows groundwater elevation (above mean sea level) at the conduit and water discharge at the spring. The dashed horizontal line represents the spring elevation. In the bottom plot, water temperature (solid lines) and specific conductivity (dashed lines) at the Conduit (black) and Spring (blue) sites are shown. Areas shaded in grey indicate low-flow periods. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

N bound to conduit sediment, or the mineralization of DON. Net-production of NO_3^- from NH_4^+ and DON is plausible as past research has indicated that the subsurface conduit water column is relatively well-oxygenated ($\text{DO}_{\text{sat}} = 76\%$; Fig. S3) and that decomposition of organic C, which is tightly coupled to mineralization of organic N, occurs at rates comparable to those in biochemically active surface streams (Husic et al., 2017b).

Coinciding with the downstream increase in NO_3^- concentration was a decrease in $\delta^{15}\text{N}_{\text{NO}_3}$, from 7.7 to 6.8‰ ($p < 10^{-3}$) (Figure 4c,d). The shift towards lighter $\delta^{15}\text{N}_{\text{NO}_3}$ would indicate that DON, SON, and NH_4^+ with isotopically lighter $\delta^{15}\text{N}_{\text{DON}}$, $\delta^{15}\text{N}_{\text{SON}}$, and $\delta^{15}\text{N}_{\text{NH}_4}$ compositions, respectively, are oxidized to NO_3^- . The rate at which this oxidation of lighter $\delta^{15}\text{N}$ occurs appears to outpace denitrification as one would otherwise expect to see an isotopically heavier

$\delta^{15}\text{N}$ signal if NO_3^- reduction to N_2 was the dominant processes. Possible sources for isotopically lighter $\delta^{15}\text{N}_{\text{DON}}$ and $\delta^{15}\text{N}_{\text{NH}_4}$ include NH_4^+ derived from fertilizer or precipitation ($\delta^{15}\text{N}_{\text{NH}_4} = -3 \pm 7\text{‰}$) and organic N from labile detrital and algal pools ($\delta^{15}\text{N}_{\text{DON}} = 5 \pm 3\text{‰}$) (Kendall et al., 2007). The $\delta^{18}\text{O}_{\text{NO}_3}$ data varied considerably and comparison of $\delta^{18}\text{O}_{\text{NO}_3}$ at upstream and downstream sites did not show statistically significant differences ($p = 0.48$) (Figure 4e,f). The lack of an apparent trend in $\delta^{18}\text{O}_{\text{NO}_3}$ does not necessarily imply that changes to $\delta^{18}\text{O}$ are not coupled to changes in $\delta^{15}\text{N}$, but rather the imprint of any one reaction may be masked by many other simultaneous reactions (Granger and Wankel, 2016).

Lastly, the upstream $\delta^{15}\text{N}_{\text{NO}_3}$ signature suggests that N loading to the conduit is primarily derived from a combination of soil,

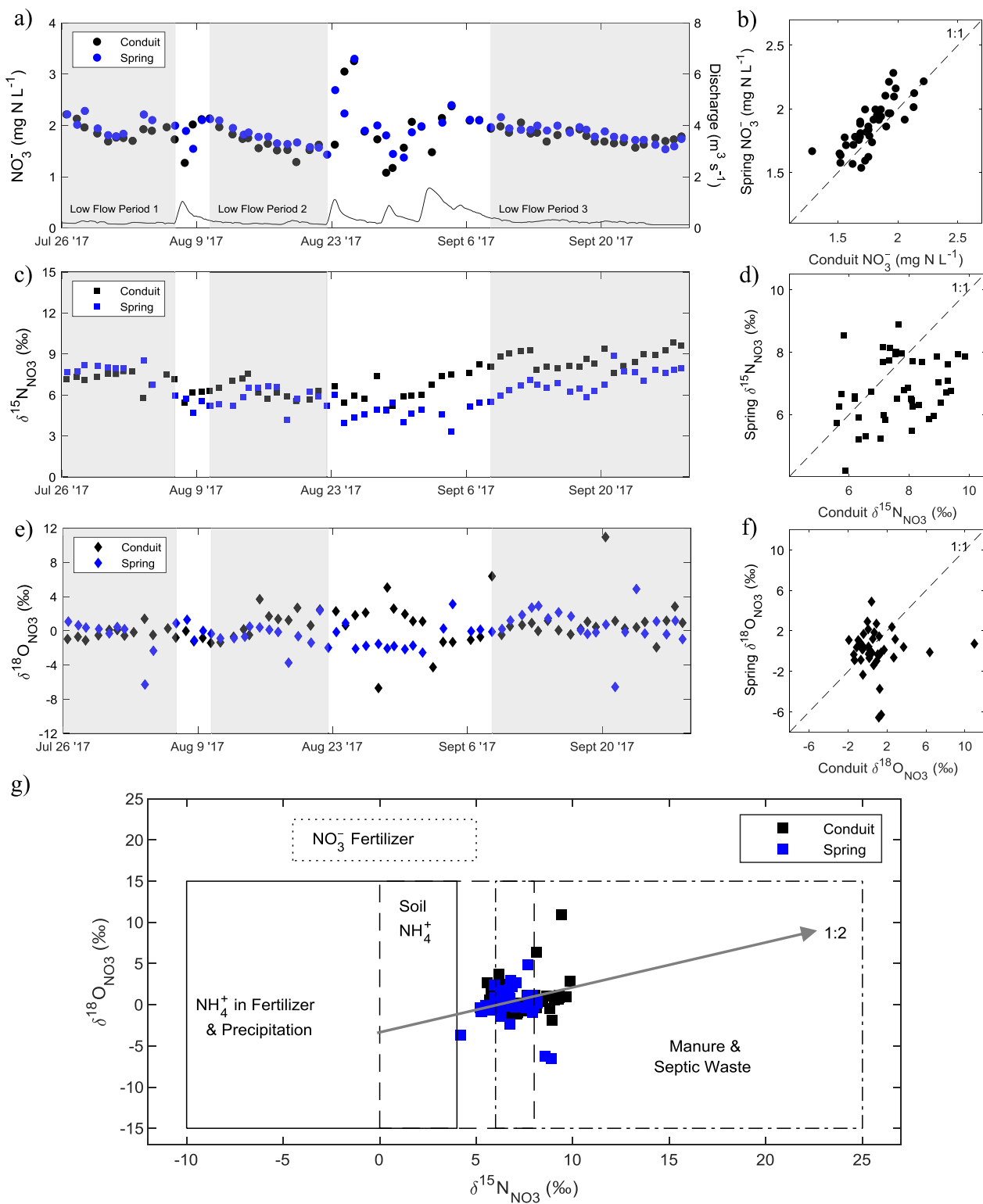


Fig. 4. Elemental and isotopic data collected at the subsurface conduit and the spring. Time-series (a, c, e) and scatter (b, d, f) plots of NO_3^- , $\delta^{15}\text{N}_{\text{NO}_3}$, and $\delta^{18}\text{O}_{\text{NO}_3}$, respectively. Only samples collected during low-flow periods are included in the scatter plots. NO_3^- concentration decreases temporally during dry periods but tends to increase longitudinally from conduit to spring ($p < 10^{-4}$). Conversely, $\delta^{15}\text{N}_{\text{NO}_3}$ increases temporally, but tends to decrease longitudinally ($p < 10^{-3}$). $\delta^{18}\text{O}_{\text{NO}_3}$ at the two sites is not significantly different ($p = 0.48$) and does not show any discernible trends. (g) $\delta^{15}\text{N}_{\text{NO}_3}$ and $\delta^{18}\text{O}_{\text{NO}_3}$ bi-plot of samples from conduit and spring locations. A grey line indicating denitrification is drawn manually through the data with a 1:2 slope. Approximate NO_3^- sources and their ranges are demarcated (adapted from Kendall et al., 2007).

fertilizer, and manure sources, with samples generally aligned with the hypothesized denitrification line (Figure 4g). As noted earlier, there is a statistically significant ($p < 10^{-3}$) shift towards lighter $\delta^{15}\text{N}_{\text{NO}_3}$ from upstream to downstream. Data results show that ambient N stable isotopes help to reflect N transformation in subsurface karst environments, with our study adding to the body of emerging stable isotope literature (Jensen et al., 2018). However, the isotope data alone are unable to differentiate between whether this new NO_3^- is sourced from allochthonous or autochthonous N. With that in mind, a numerical model provides a tool to remedy this uncertainty and to elucidate processes that are not revealed by data streams alone.

4.2. Isotope-aided numerical modeling of nitrate fate

The isotope dataset allowed for adding a second set of modeling equations without adding additional unknowns, which helped constrain uncertainty and equifinality in model parameters (Fig. 5 and Table 1). After evaluating the numerical model results against the three elemental criteria – NSE_{NO_3} , R_{NH_4} , and R_{DON} (see Fig. 2) – the posterior distribution of several parameters was reduced significantly from the assumed prior distribution (Figure 5a and Table 1). Subsequently, when integrating the isotopic criteria – $\text{NSE}_{\delta^{15}\text{N}}$ – the posterior distributions of the denitrification and nitrification rate constants were further constrained as was NH_4^+ recharge to the system (Figure 5b). Reaction rates and concentrations of the inorganic N phases (NO_3^- and NH_4^+) are relatively sensitive to model calibration (see ϵ_{nitr} , ϵ_{den} , k_{nitr} , β_{den} , $C_{\text{NH}_4\text{-rec}}$, and $\delta^{15}\text{N}_{\text{NH}_4\text{-rec}}$) whereas rates and concentrations that impact the organic N phases (DON and SON) are not as sensitive (see ϵ_{imm} , k_{min} , $C_{\text{DON-rec}}$, and $\delta^{15}\text{N}_{\text{DON-rec}}$ in Fig. 5). The lack of sensitivity in the organic N-related parameters likely arises from the slower processing rates and more recalcitrant nature of DON relative to NH_4^+ and NO_3^- . Isotopic fractionation parameters related to nitrification (ϵ_{nitr}) and denitrification (ϵ_{den}) were on the lower range reported in the literature with posterior medians of +3.8 and +2.5‰, respectively (Figure 5b). The low values of ϵ_{nitr} and ϵ_{den} indicates that although nitrification and denitrification may occur in the

subsurface, these processes do not substantially fractionate the isotopic signature of their respective pools. These results tend to agree with a study by Heffernan et al. (2012), where small ϵ_{den} values of between 5.3 and 7.5 were estimated using dissolved gas data at 61 karst springs. In non-karst systems, reported values of ϵ_{nitr} and ϵ_{den} can range considerably from +1 to +26‰ and +1 to +18‰, respectively, and can be influenced by nutrient, enzyme, and diffusion limitations (Kendall et al., 2007 and references within) and whether denitrification is occurring in the water column (high enrichment) or within the benthic zone (low enrichment) (Dähnke and Thamdrup, 2013).

Nitrate fate and transport model results agreed well with biogeochemical data observations at Royal Spring and indicate that the source of the additional NO_3^- was from autochthonous production (i.e. the conversion of SON, DON, and NH_4^+ to NO_3^-) rather than external, allochthonous inputs (Figure 6a). The NSE statistics for the optimal NO_3^- model run were 0.39 and 0.28 for calibration and validation periods, respectively. Model results are consistent with data and show longitudinal increases in NO_3^- concentration, brought on by the oxidation of DON and NH_4^+ , from the Phreatic Conduit to Royal Spring. The NSE statistics for the optimal $\delta^{15}\text{N}_{\text{NO}_3}$ model run were 0.27 and 0.15 for calibration and validation periods, respectively (Figure 6b). These results are satisfactory given the complexity and difficulty of modeling isotopic N fate at the spatial and temporal resolution performed in this study. Modeled $\delta^{15}\text{N}_{\text{NO}_3}$ results are consistent with data and indicate that NO_3^- becomes isotopically depleted in ^{15}N during longitudinal transport from input at the upstream Phreatic Conduit to discharge at the terminal Royal Spring. The model was generally not-sensitive to the 6% lateral inflow from the Phreatic Conduit to Royal Spring as the quantity of the inflow to the total load was relatively small and the concentration of material was similar to existing, in-conduit material. Some disagreement between the calibrated model and data occurs when the model overestimates $\delta^{15}\text{N}_{\text{NO}_3}$ at the beginning of “Low Flow Period 3”. This is likely associated with a process not captured by our spatially and temporally integrated model structure, such as the occurrence of non-stationary ‘hot moments’ or ‘hot spots’ that can disproportionately process matter within the

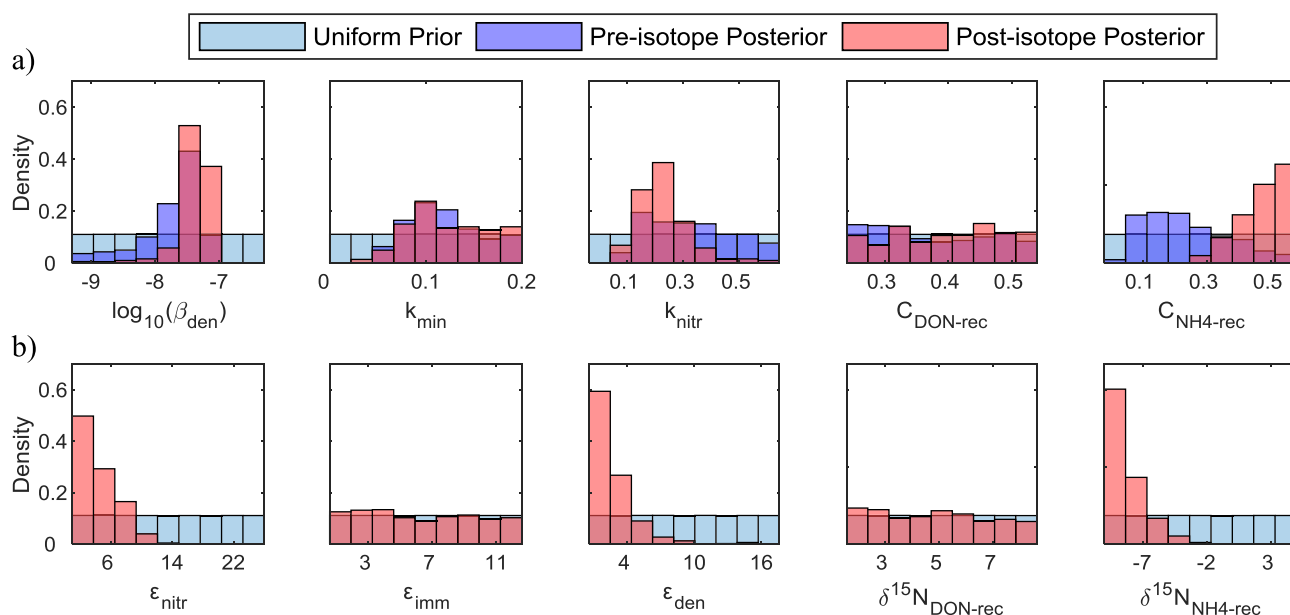


Fig. 5. Prior and posterior distributions of model parameters and inputs for the (a) elemental and (b) isotopic models. Note: only the elemental model parameters have both pre-isotope and post-isotope distributions.

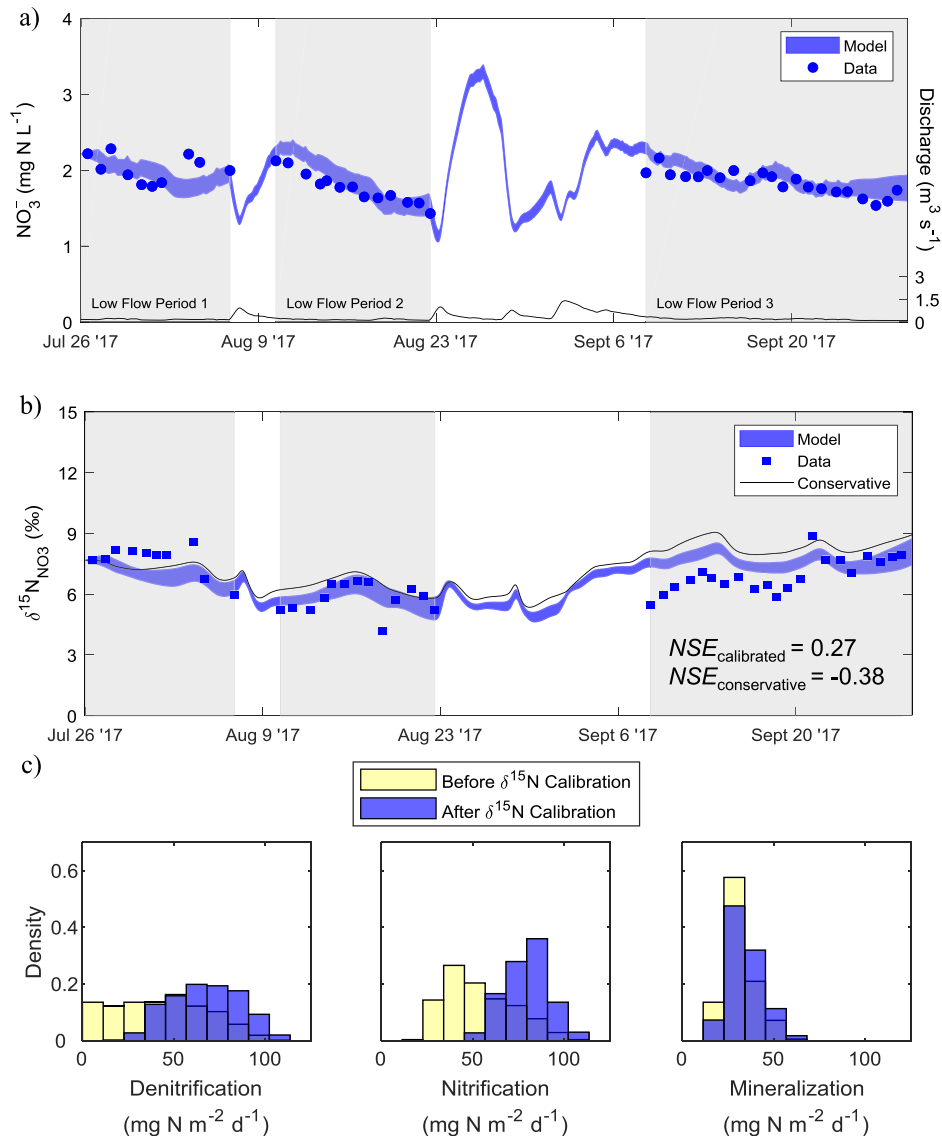


Fig. 6. Modeling results of (a) NO_3^- and (b) $\delta^{15}\text{N}_{\text{NO}_3}$ for three low-flow periods contained within the study duration. Spring output is presented as the 95% prediction bound from the set of acceptable model simulations. Blue dots represent discrete data points collected at the primary spring. The grey line in (b) represents model outputs if conservative transport is assumed (i.e. no reactivity of N within the conduit). (c) Observed changes in modeled nitrification, denitrification, and mineralization rates *before* and *after* integrating isotopic data. Isotope data reveals a greater degree of biogeochemical processing not observed with elemental data streams alone. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

time and space domains, respectively (Krause et al., 2017). For example, large spatial heterogeneity in the release of NH_4^+ or NO_3^- from the karst bed could impact predictions. This explanation is plausible as the event preceding Low Flow Period 3 was the largest in the two-month study and significant storm flow has been shown to disturb the structure of the biologically active surficial layer of the benthos, which can expose previously buried material to processing (Ford et al., 2015).

Numerical modeling of isotope dynamics allowed for the estimation of isotopic overprinting of nitrification on denitrification. Results from testing multiple scenarios (i.e., no denitrification, no nitrification, and conservative transport) indicate that if either nitrification or denitrification are excluded from the modeling framework, NO_3^- and $\delta^{15}\text{N}_{\text{NO}_3}$ results are not accurately represented (Fig. S4). In the case of no denitrification, NO_3^- is consistently over-estimated ($NSE_{\text{NO}_3} = -0.36$). In the case of no nitrification, the $\delta^{15}\text{N}_{\text{NO}_3}$ signature is greatly over-estimated

($NSE_{\delta^{15}\text{N}} = -0.73$). Lastly, with the conservative transport assumption (i.e. all reaction rates set to “0”), the model does not satisfactorily simulate the $\delta^{15}\text{N}_{\text{NO}_3}$ signature ($NSE_{\delta^{15}\text{N}} = -0.38$). The integration of both elemental (NO_3^-) and isotopic ($\delta^{15}\text{N}_{\text{NO}_3}$) datasets was crucial to determining the existence and relative importance of in-conduit processes. While our results indicates that denitrification does occur during transport, its signal is not as pronounced due to concurrent NO_3^- production by nitrification. Thus, our numerical model allowed for the assessment of isotopic overprinting, which would have otherwise masked in-conduit denitrification.

Isotope-aided numerical modeling also helped to reduce uncertainty regarding the dominant processes controlling NO_3^- fate in subsurface karst. In general, we found NO_3^- elemental modeling alone – that is, without the integration of $\delta^{15}\text{N}_{\text{NO}_3}$ data – could accurately represent NO_3^- concentration at the spring, but would under-represent internal processing rates. The additional $\delta^{15}\text{N}_{\text{NO}_3}$

data series revealed that the subsurface karst facilitates transformations at a greater rate than indicated by elemental NO_3^- data alone. Although the increased nitrification and denitrification rates offset one another with regards to net- NO_3^- concentration, there is a non-linear effect of fractionation on the $\delta^{15}\text{N}_{\text{NO}_3}$ signature, which provides the means for more accurately calibrating reaction rates. Pre-isotope calibration rates for denitrification, nitrification, and mineralization were $37.6 (\pm 25.0)$, $50.3 (\pm 17.9)$, and $29.7 (\pm 9.8)$ $\text{mg N m}^{-2} \text{d}^{-1}$, respectively. On the other hand, post-isotope calibration showed elevated biogeochemical processing with denitrification, nitrification, and mineralization rates increasing to $67.0 (\pm 19.0)$, $79.5 (\pm 12.1)$, and $34.1 (\pm 9.3)$ $\text{mg N m}^{-2} \text{d}^{-1}$, respectively. Immobilization rates were an order-of-magnitude lower (4.67 ± 0.01 $\text{mg N m}^{-2} \text{d}^{-1}$) and were unaffected by the isotopic calibration. The increase in the estimates of denitrification and nitrification are significant ($p < 10^{-4}$) and represent a 78 and 58% increase in processing over pre-isotope estimates, respectively, and the variance in all rate estimates is reduced. Ambient N isotopes have shown success when applied to constrain NO_3^- fate in surface systems (Ford et al., 2017; Kaown et al., 2009), and our results complement this growing body of literature and show the potential of ambient N isotopes to assist with N modeling in karst groundwater.

4.3. Nitrate removal in sinking streams

The karst conduit's NO_3^- removal via denitrification (67.0 ± 19.0 $\text{mg N m}^{-2} \text{d}^{-1}$) falls between the bounds of groundwater and surface water systems (Fig. 7 and Table 2). Nitrate removal in streams, rivers, and creeks varies significantly ($100\text{--}1000$ $\text{mg N m}^{-2} \text{d}^{-1}$) and is typically greater than other ecosystems as rivers are characterized by high biochemical gradients of oxygen, nutrients, and organic matter (Arango et al., 2007; Piña-Ochoa and Álvarez-Cobelas, 2006). On the other hand, groundwater ($1\text{--}10$ $\text{mg N m}^{-2} \text{d}^{-1}$) and soil ($10\text{--}100$ $\text{mg N m}^{-2} \text{d}^{-1}$) zones are characterized by lesser denitrification fluxes than streams, rivers, and creeks, but constitute a greater proportion of watershed land area and have longer residence times (Anderson

et al., 2014; Jahangir et al., 2012; Seitzinger et al., 2006). In comparison to other karst systems, average aquifer denitrification up-gradient of springs ($0.3\text{--}3$ $\text{mg N m}^{-2} \text{d}^{-1}$) and within turloughs ($5\text{--}50$ $\text{mg N m}^{-2} \text{d}^{-1}$) falls within the ranges of removal found in equivalent non-karst systems (Heffernan et al., 2012; Husic et al., 2019b; McCormack et al., 2016; Piña-Ochoa and Álvarez-Cobelas, 2006). Thus, karst conduits act to remove NO_3^- at greater rates than karst and non-karst soil, lake, and groundwater systems, but at lower rates than surface streams, primarily due to a lack of a consistent labile (e.g., autotrophic) carbon source (Zeng et al., 2019). Our estimation of NO_3^- fate in karst conduits, using stable isotopes and numerical modeling, contextualizes the potential role and importance of conduit sediment to remove NO_3^- from the broader nutrient cascade.

Nitrification in the sinking stream of our study (79.5 ± 12.1 $\text{mg N m}^{-2} \text{d}^{-1}$) was on the higher end of rates reported in the literature, which is consistent with periods of high NH_4^+ levels observed in our system. In the Royal Spring basin, the long-term average concentration of NH_4^+ recharging the subsurface is 0.12 mg N L^{-1} while the concentration discharging at the spring is 0.07 mg N L^{-1} (Fig. S1). However, shortly prior to and during the study period, high concentrations of NH_4^+ were detected at Royal Spring (e.g., May 5, 1.2 mg N L^{-1} ; June 12, 0.38 mg N L^{-1} , and Sept. 15, 0.40 mg N L^{-1}), and in some instances, such as May 5 and June 1, the water treatment plant at Royal Spring shut down operations due to high NH_4^+ concentrations (personal communication, Georgetown Municipal and Water Sewer Service, 2017). Likewise, long-term DON concentrations decrease from 0.35 mg N L^{-1} for recharge to 0.23 mg N L^{-1} at the spring (Fig. S1), which may provide an additional source of NH_4^+ through mineralization of the organic N, as has been indicated recently in other karst systems (e.g., Musgrove et al., 2016). Our results are consistent with other studies of agriculture-impacted watersheds where maximum rates of nitrification can be up to tenfold greater than denitrification (Kemp and Dodds, 2002) and net-nitrifying streams are associated with higher NO_3^- concentrations. Prior studies have suggested the importance of nitrification and denitrification within the phreatic karst zone (e.g., Einsiedl and Mayer, 2006; Musgrove et al., 2016), and our study

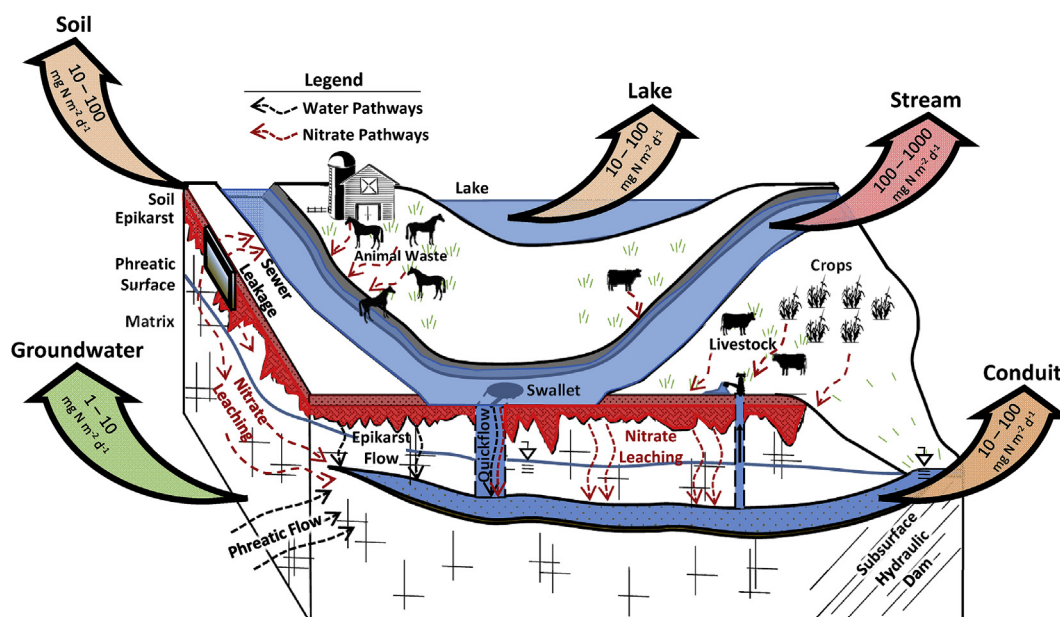


Fig. 7. Schematic highlighting the denitrification rates of various pathways, including lake, stream, groundwater, soil, and conduit systems. The number inset within the arrows is the typical range of rates (see Table 2 for references). Figure adapted from Husic et al., 2019b.

Table 2
Denitrification rate comparison across surface, soil, groundwater, and karst ecosystems.

Study Location(s)	Waterbody (number of sites/estimates)	Denitrification (mg N m ⁻² d ⁻¹)	Reference
<i>Surface Water</i>			
Midwestern USA	Agricultural streams (n = 8)	132.6 (±114.5)	Arango et al. (2007) and refs within
Global	Streams from diverse biomes (n = 23)	274.9 (±652.2)	
USA & Europe	Riverbed sediment (n = 21)	185.9 (±239.3)	Piña-Ochoa and Álvarez-Cobelas (2006) and refs within
Global	Lakebed sediment (n = 17)	35.1 (±35.0)	
<i>Soil and Groundwater</i>			
USA & Europe	Riparian soil in agricultural landscapes (n = 10)	53.0 (±66.8)	Anderson et al. (2014)
Global	Watershed soil denitrification (n = 9)	27.8 (±32.5)	Seitzinger et al. (2006)
Ireland	Agricultural groundwater removal (n = 8)	2.1 (±1.6)	Jahangir et al. (2012)
Global	Watershed groundwater denitrification (n = 9)	1.2 (±1.2)	Seitzinger et al. (2006)
<i>Karst Landscapes</i>			
Ireland	Karst lakes (turloughs) (n = 2)	25.6 (±26.6)	McCormack et al. (2016)
Florida USA	Flow-weighted aquifer rate (n = 61)	0.3 (±N/A)	Heffernan et al. (2012)
Kentucky USA	Areal aquifer rate (n = 1)	2.2 (±1.1)	Husic et al. (2019b)
Kentucky USA	Conduit bed sediment (n = 1)	67.0 (±19.0)	This study

provides a quantification of its importance in the context of karst conduits.

Given the comparatively high transformations rates indicated by stable isotopes and numerical modeling, the net role of karst conduits in N cycling appears to be non-conservative, countering the paradigm of karst conduits as conservative conveyors of energy and nutrients. The mechanism for this non-conservative behavior is the presence of a terrestrially derived, organically rich sediment substrate, which acts as an interface for NO₃⁻ removal. In surface rivers, these benthic interfaces are sometimes considered ecohydrological hot spots and control points where significant processing of organic matter and nutrients occurs across steep biochemical gradients (Bernhardt et al., 2017; Krause et al., 2017). Thus, it is reasonable that the NO₃⁻ removal rate in karst conduits (67.0 ± 19.0 mg N m⁻² d⁻¹) falls between surface (100–1000 mg N m⁻² d⁻¹) and subsurface (1–10 mg N m⁻² d⁻¹) systems as conduits act as an intermediary pathway that mediate terrestrial loading of riverine organic matter with subterranean leaching of nutrients. Further support for the relative removal efficiency of karst is provided in a study by Piña-Ochoa and Álvarez-Cobelas (2006) analyzing the qualitative factors influencing denitrification (i.e. light conditions, presence of plants, oxygen levels, organic C quality, total phosphorus concentrations, and nitrate levels). Piña-Ochoa and Álvarez-Cobelas (2006) conclude that low light conditions, the absence of plants, low oxygen within anaerobic sediment pockets, low phosphorus, high organic C, and high nitrate are correlated with increased denitrification rates. The present karst systems fits all but the phosphorus criteria. Given the combination of factors that include buffered loading to the conduit, mixing with the sediment substrate, and high background aquifer NO₃⁻ levels it is highly plausible that the karst conduit is an efficient denitrifying system. To conclude, karst conduit pathways that are well-recharged by surface-derived organic material are likely non-conservative and should be considered as such when accounting for N fate and constructing N budgets.

5. Conclusions

- 1) The karst sinking stream's NO₃⁻ removal falls between the bounds of groundwater systems and surface water systems, providing a counterpoint to the paradigm that karst conduits are conservative source-to-sink conveyors.
- 2) Nitrogen concentration and isotope data results show that the karst conduit acts as a net source of NO₃⁻ via nitrification of soil organic N and fertilizer, adding to the emerging body of ambient N isotope literature for estimating N fate.

- 3) The isotope-aided numerical model reduced uncertainty when estimating NO₃⁻ removal and quantified isotopic overprinting for the karst sinking stream, adding to the growing body of literature on the usefulness of isotope-aided methods to assist with stream and watershed water-quality modeling.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.watres.2019.115348>.

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